Engineering Chemistry

UNIT – 4 : Polymer Chemistry

From monomers to materials — how everyday plastics, rubbers and smart polymers are built, with clear repeat-unit sketches.

Topics in this unit

1. Polymers, functionality & polymerization
2. Chain, step & coordination polymerization
3. Plastics: PVC, Teflon, Bakelite, Nylon-6,6, carbon fibres
4. Elastomers: Buna-S & Buna-N
5. Conducting polymers
6. Biodegradable polymers (PGA, PLA)

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. What is a Polymer?

Look around you — bottles, pipes, clothes, tyres, phone cases. Almost all of them are made of polymers. A polymer is a very large molecule built by joining hundreds or thousands of small, repeating units together like beads on a string.

DEFINITION

Polymer: a large molecule (a macromolecule) made of many small repeating units called monomers, joined together by strong covalent bonds.

DEFINITION

Polymerization: the chemical process that joins many monomer molecules into a polymer chain.

A few words you will meet again and again:

- **Monomer** — the small starting molecule (e.g. ethene, $CH_2=CH_2$).
- **Repeat unit** — the part that repeats over and over inside the chain.
- **Degree of polymerization (n)** — the number of repeat units in one chain. A big n means a long, heavy chain.

$$n\ CH_2=CH_2 \rightarrow -(CH_2-CH_2)_n-$$

n molecules of ethene join to give polyethylene (polythene)

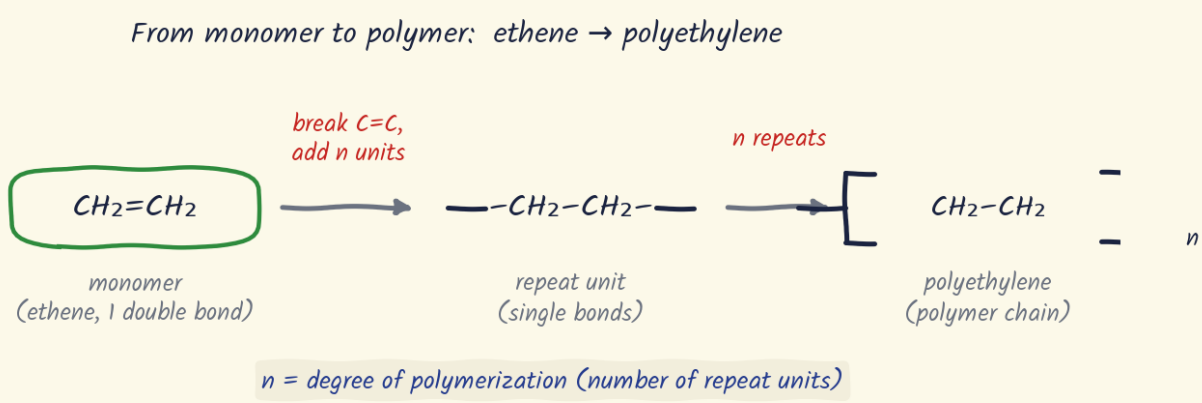


Fig 1 — One monomer opens its double bond, becomes a repeat unit, and n of them link into a long polymer chain.

EXAMPLE

Polyethylene: n molecules of ethene ($CH_2=CH_2$) join end-to-end to form $-(CH_2-CH_2)_n-$ — the plastic used in carry-bags and bottles.

NOTE

Natural polymers also exist — **starch**, **cellulose**, **proteins** and **DNA** are all polymers made by nature. In this unit we focus mainly on synthetic (man-made) polymers.

2. Functionality of a Monomer

Why do some polymers form soft, flexible threads while others set into hard, rigid solids? The answer is **functionality** — how many bonding arms each monomer has.

DEFINITION

Functionality: the number of **reactive sites** (bonding points) on a monomer that can join to other monomers.

The functionality decides the shape of the final polymer:

- Functionality = 2 → each monomer links to just two neighbours → a straight linear chain. Example: ethylene, adipic acid.
- Functionality ≥ 3 → monomers link in several directions → a branched or cross-linked network. Example: glycerol; phenol + formaldehyde.

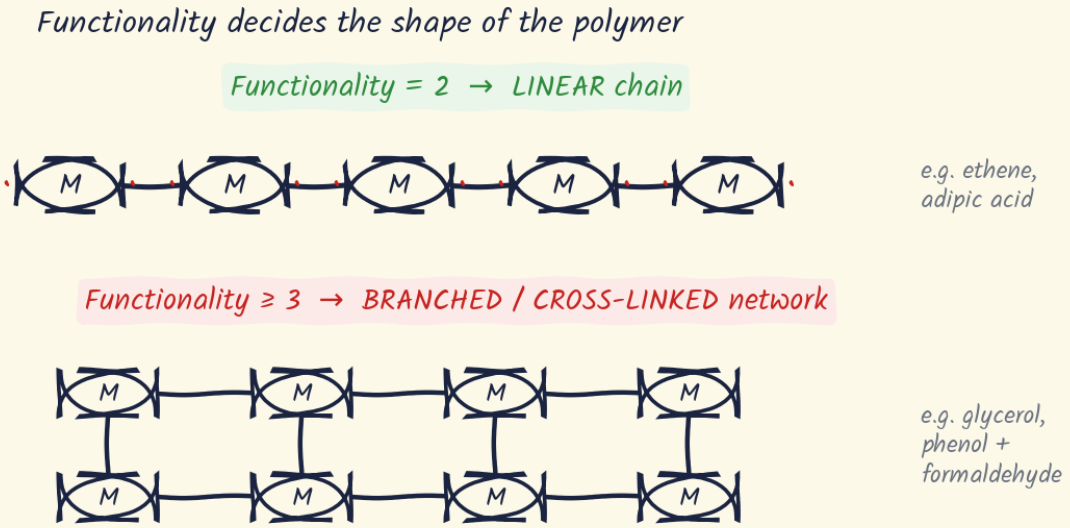


Fig 2 — Bifunctional monomers give a linear chain; tri- (or higher) functional monomers build a cross-linked network.

EXAMPLE

Ethene has functionality 2 (one double bond opens into two bonds) → linear polythene. Phenol + formaldehyde gives functionality ≥ 3 → the rigid 3-D network of Bakelite.

KEY POINT

Simple rule: 2 arms → thread, 3 or more arms → net. Linear polymers are usually soft and remouldable; cross-linked ones are hard and permanent.

3. Chain-Growth (Addition) Polymerization

There are three main ways to build a polymer. The first is **chain-growth**, also called **addition** polymerization.

DEFINITION

Chain-growth polymerization: unsaturated monomers (with a C=C double bond) add on rapidly, one after another, through a reactive intermediate. **No small molecule is lost**, and a high molecular weight is reached very quickly.

- Needs monomers with a **double bond** (C=C), e.g. ethene, vinyl chloride.
- The chain grows **very fast** through a reactive centre (a free radical).
- **No by-product** is formed — every atom of the monomer ends up in the polymer.

Free-radical mechanism — three steps

- **Initiation** — an initiator (e.g. **benzoyl peroxide**) breaks into free radicals (R•) that attack a monomer.
- **Propagation** — the radical adds monomer after monomer, growing the chain rapidly.
- **Termination** — two growing chains join (**combination**) or swap an atom (**disproportionation**), and growth stops.

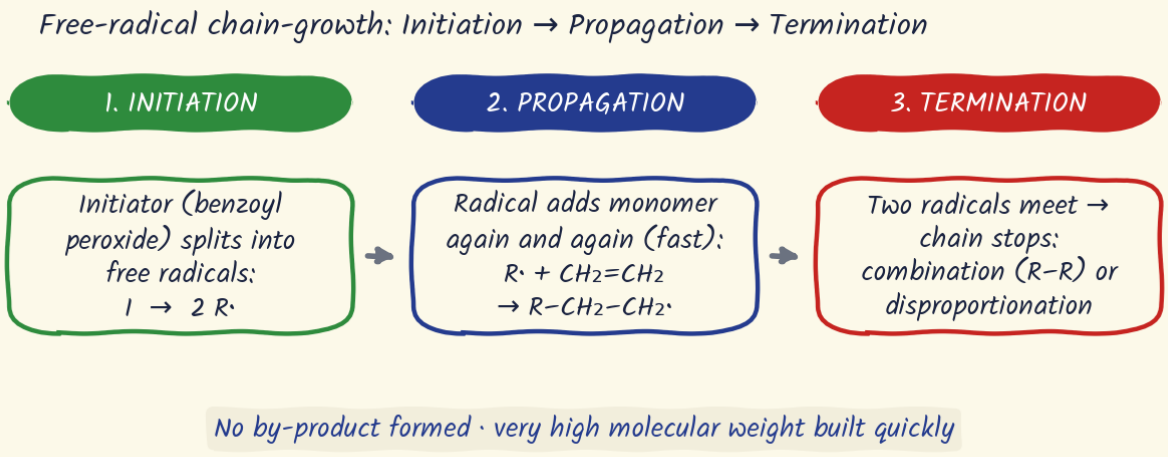


Fig 3 — The three steps of free-radical chain-growth polymerization.

EXAMPLE

Polyethylene is made this way: ethene ($CH_2=CH_2$) molecules add on one by one to give long $-(CH_2-CH_2)_n-$ chains.

4. Step-Growth (Condensation) Polymerization

The second route is step-growth, usually called condensation polymerization. Here growth is slow and steady, and a small molecule is thrown out at each step.

DEFINITION

Step-growth polymerization: bi-functional monomers react step by step; at each joint a small molecule (usually H_2O or HCl) is eliminated. Molecular weight builds up slowly.

- Monomers carry two reactive groups (e.g. –COOH, –OH, $-NH_2$).
- A small molecule is released at every linkage (condensation).
- Any two units can join (monomer+monomer, dimer+trimer…), so the chain grows gradually.

EXAMPLE

Nylon-6,6 (adipic acid + hexamethylenediamine), PET polyester and Bakelite (phenol + formaldehyde) are all made by condensation.

Chain-growth vs step-growth

Feature	Chain-growth (addition)	Step-growth (condensation)
Monomer	has a C=C double bond	bi-functional (–OH, –COOH, $–NH_2$)
Growth	very fast, adds one monomer at a time	slow, any two units combine
By-product	none	small molecule (H_2O / HCl)
Mol. weight	high almost immediately	builds up gradually
Examples	polythene, PVC, Teflon	Nylon-6,6, PET, Bakelite

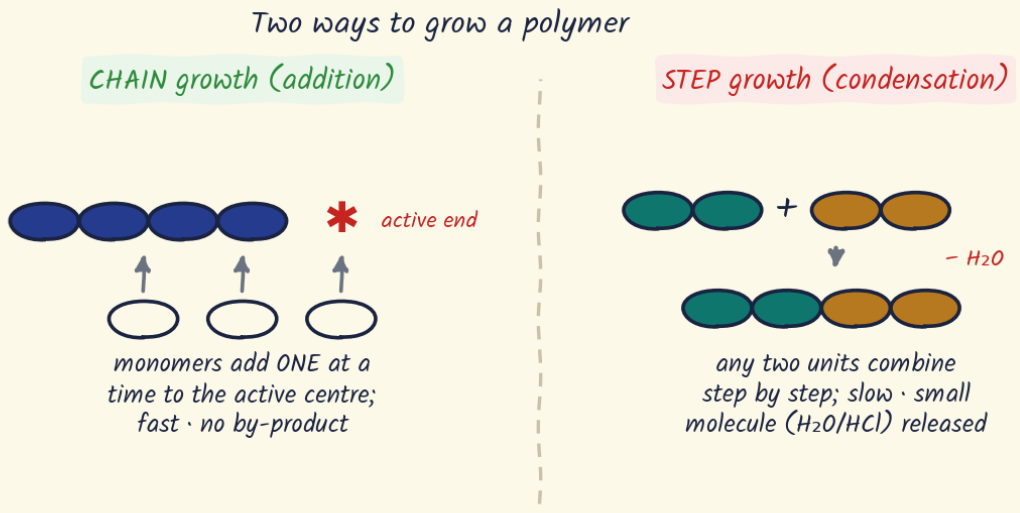


Fig 4 — Chain-growth adds single monomers to one active end; step-growth joins any two units and releases a small molecule.

5. Coordination Polymerization (Ziegler–Natta)

The third route uses a special catalyst to build very neat, well-ordered chains. It gave us modern high-quality plastics and won a Nobel Prize.

DEFINITION

Coordination polymerization: polymerization carried out with a **Ziegler–Natta catalyst** — $TiCl_4$ + $Al(C_2H_5)_3$ (triethylaluminium). It gives **stereoregular** (isotactic), linear, high-density polymers, and works at **low pressure**.

- Monomers insert one by one at the metal centre, so each unit lines up the same way.
- The product is **stereoregular (isotactic)** — side groups all on the same side.
- Chains are **linear and high-density**, giving stronger, denser plastics.

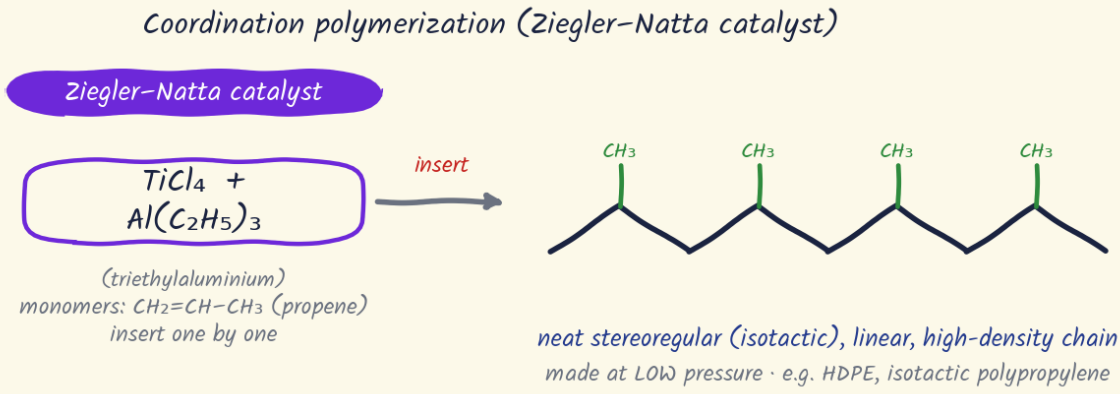


Fig 5 — A Ziegler–Natta catalyst lines monomers up into a neat, stereoregular, linear chain.

EXAMPLE

HDPE (high-density polyethylene) and **isotactic polypropylene** are made by coordination polymerization.

KEY POINT

Big advantage: the catalyst gives control over structure and stereochemistry, so the plastic is denser, more crystalline and much stronger.

6. Plastics: Thermoplastic vs Thermosetting

Plastics are polymers that can be moulded into shapes. They fall into two big families based on how they behave when heated.

DEFINITION

Thermoplastics: soften on heating and can be remoulded again and again. They have linear or branched chains held by weak forces. e.g. PVC, Teflon, polythene, nylon.

DEFINITION

Thermosetting plastics: set permanently hard on heating (a 3-D cross-linked network) and cannot be remoulded. e.g. Bakelite, epoxy.

Property	Thermoplastic	Thermosetting
Structure	linear/branched chains, weak forces	3-D cross-linked network
On heating	softens → can remould	sets hard → cannot remould
Reusable?	yes, recyclable	no
Examples	PVC, Teflon, polythene, nylon	Bakelite, epoxy resin

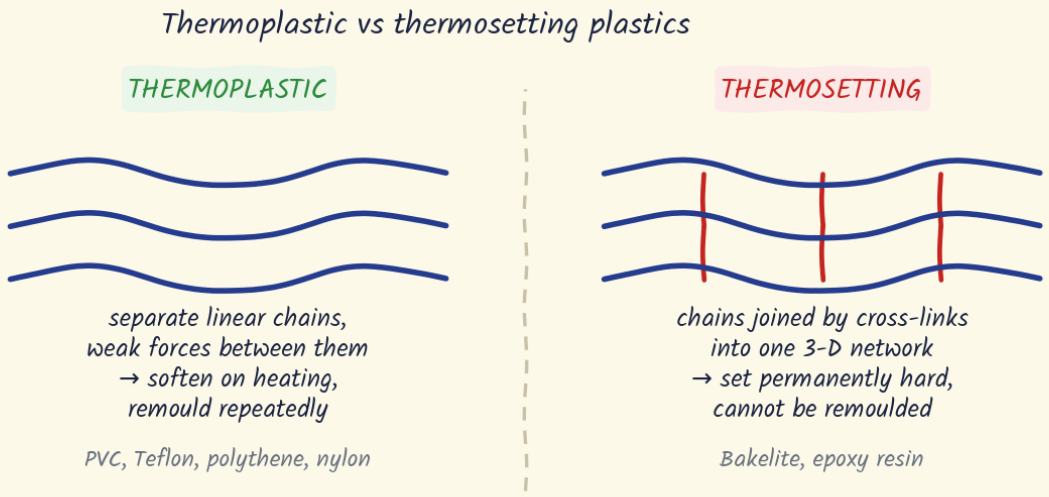


Fig 6 — Thermoplastics are separate chains that slide when heated; thermosets are one locked cross-linked network.

KEY POINT

Memory hook: thermo-PLASTIC stays plastic (re-softens); thermo-SET is set for good.

7. Important Plastics

Let us meet five everyday plastics, their monomers and where they are used.

PVC — Polyvinyl Chloride

Made by addition polymerization of vinyl chloride ($CH_2=CHCl$). It can be made rigid, or soft and flexible by adding a plasticizer.

- Uses: pipes, cable insulation, raincoats, flooring, bottles.

Teflon (PTFE)

Addition polymer of tetrafluoroethylene ($CF_2=CF_2$). The strong C–F bonds make it chemically inert, heat-resistant, non-stick and a good insulator

with very low friction.

- Uses: non-stick cookware coatings, gaskets and seals, electrical insulation.

Bakelite — Phenol-Formaldehyde

A thermosetting plastic made by condensation of phenol + formaldehyde into a 3-D cross-linked network. It is hard and resists both heat and electricity.

- Uses: electrical switches, plugs, handles, fittings.

Nylon-6,6 — a Polyamide

Made by condensation of adipic acid (6 carbons) + hexamethylenediamine (6 carbons) — hence the name 6,6. The chains are joined by amide linkages (–CO–NH–), so nylon is a polyamide. It is strong, elastic and abrasion-resistant.

- Uses: fibres, ropes, textiles, gears, bearings, carpets.

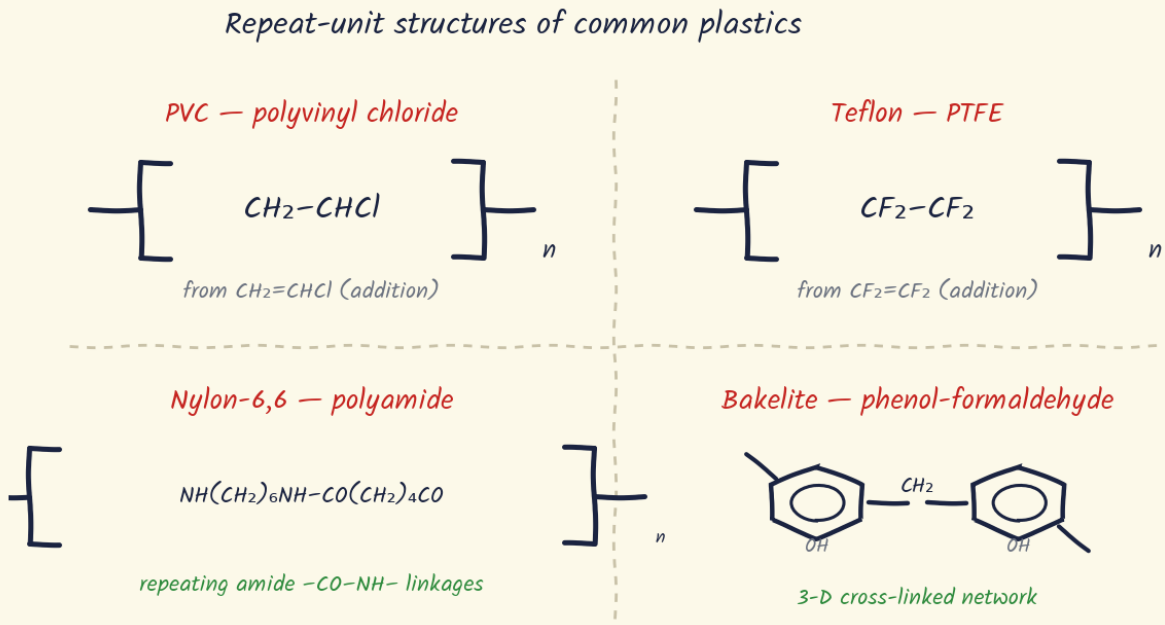


Fig 7 — Repeat-unit sketches: PVC, Teflon, Nylon-6,6 (note the –CO–NH– amide links) and the Bakelite network.

Carbon Fibres

Fibres of at least **90% carbon** made by the controlled **pyrolysis** (heating) of **PAN — polyacrylonitrile**. They have a very high strength-to-weight ratio, high stiffness and resist heat.

- **Uses:** aerospace parts, sports goods, automobiles, construction.

> EXAMPLE
> Carbon fibre is **lighter than aluminium but stiffer than steel** — which is why aircraft, racing cars and premium bicycles use it.

8. Elastomers: Buna-S and Buna-N

Elastomers are rubber-like polymers that stretch a lot and then snap back to shape. Two important synthetic rubbers are Buna-S and Buna-N — both **copolymers** (made from two different monomers) of

butadiene.

Buna-S (SBR)

A copolymer of butadiene + styrene, made by free-radical emulsion polymerization. It is tough and abrasion-resistant.

- Uses: automobile tyres, shoe soles, cable insulation, conveyor belts.

Buna-N (NBR)

A copolymer of butadiene + acrylonitrile. The –C≡N groups make it oil-, solvent- and heat-resistant.

- Uses: fuel hoses, oil seals, gaskets, protective gloves.

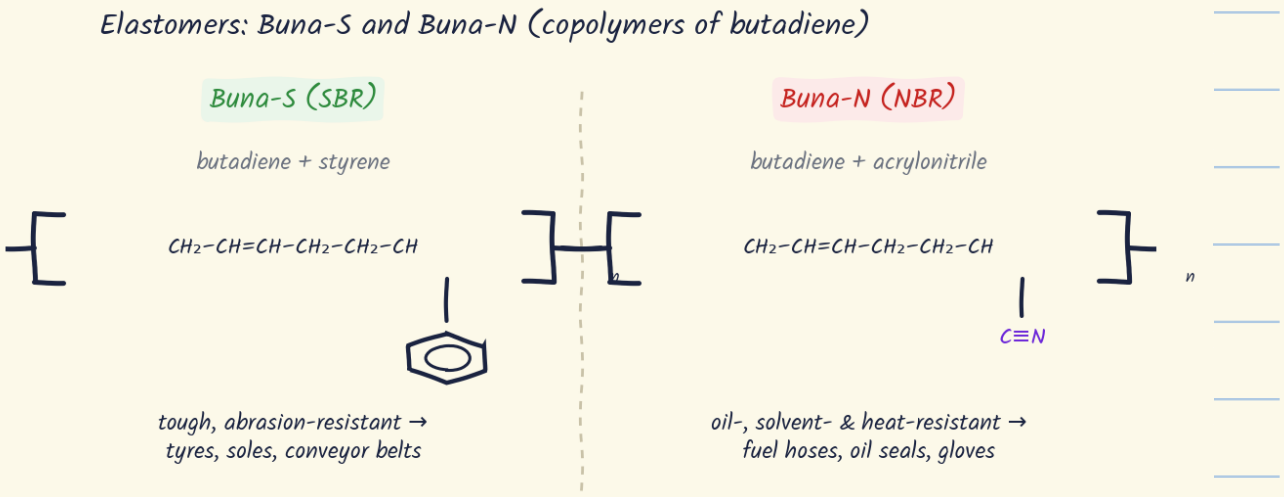


Fig 8 — Repeat-unit sketches of Buna-S (butadiene + styrene) and Buna-N (butadiene + acrylonitrile).

> EXAMPLE
>
> The name Buna comes from Butadiene + Na (sodium, the old catalyst). Buna-S has Styrene; Buna-N has the Nitrile group.

9. Conducting Polymers

Most plastics are insulators — but some special polymers can actually conduct electricity almost like a metal. These are the conducting polymers.

DEFINITION

Conducting polymer: a polymer that conducts electricity because of a conjugated backbone (alternating single and double bonds → delocalized π electrons) plus doping.

- Polyacetylene $(-CH=CH-)_n$ — doping with I_2 (oxidation, p-type) or Na (reduction, n-type) makes it highly conducting.
- Polyaniline — its conductivity is controlled by its oxidation state and protonation (acid doping).

How does it conduct?

The conjugated backbone gives a delocalized π pathway along the chain. Doping then removes or adds electrons, creating mobile charge carriers (polarons) that drift along the chain under an electric field → current flows.

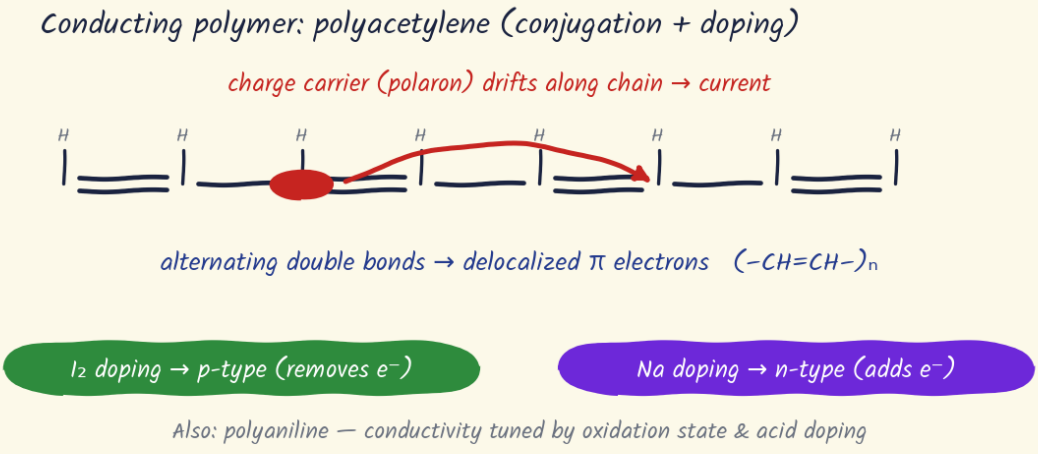


Fig 9 — A conjugated polyacetylene chain; doping creates a mobile charge carrier that drifts along the π system.

Applications:

- Rechargeable batteries and electrodes.
- Sensors and antistatic coatings.
- LEDs / OLEDs and electrochromic displays.
- EMI (electromagnetic interference) shielding.

KEY POINT

Two things are needed together: a conjugated backbone (the highway) and doping (the moving charges). Remove either and conduction stops.

10. Biodegradable Polymers

Ordinary plastics last for centuries and choke the environment. Biodegradable polymers are designed to break down naturally — a greener choice.

DEFINITION

Biodegradable polymer: a polymer that micro-organisms can break down into CO_2, H_2O and biomass, so it does not pile up as waste.

- PGA (polyglycolic acid) — made from glycolic acid by condensation. Used for dissolvable surgical sutures and tissue scaffolds.
- PLA (polylactic acid) — made from lactic acid (from corn starch / sugar fermentation). Used for packaging, cups, cutlery, 3-D printing filament, sutures, drug delivery and implants.

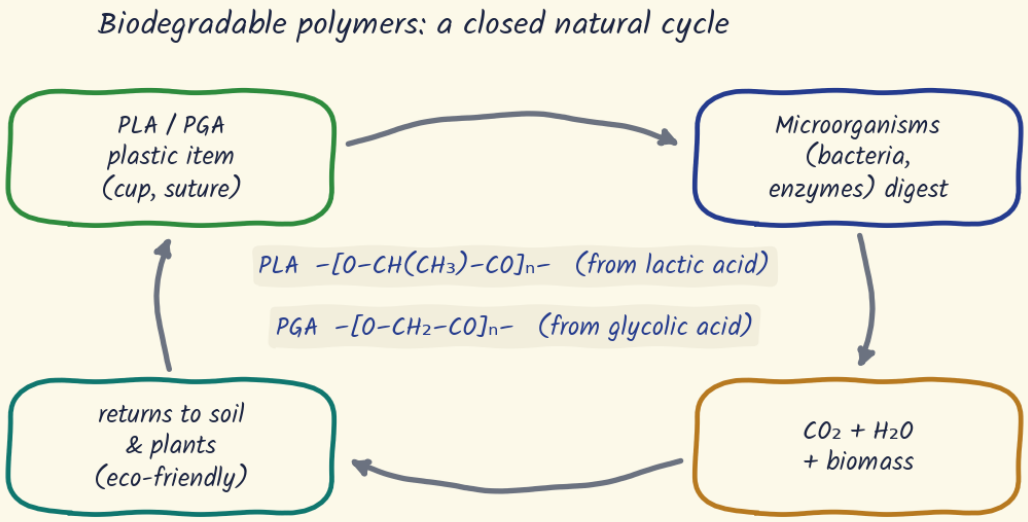


Fig 10 — A biodegradable polymer item is digested by micro-organisms into CO_2, water and biomass, returning to nature.

EXAMPLE

A PLA cup thrown into a compost heap is eaten by microbes and turns into CO_2, water and biomass — unlike a normal plastic cup that survives for hundreds of years.

II. Applications & Importance

Polymers are everywhere in modern engineering and daily life:

- **Packaging & daily use:** polythene bags, PVC bottles, PET water bottles.
- **Engineering parts:** nylon gears and bearings, HDPE pipes, carbon-fibre aerospace and automobile components.
- **Electrical insulation:** Bakelite switches, PVC cable sheathing, Teflon insulation.
- **Tyres & sealing:** Buna-S tyres and belts; Buna-N oil seals and fuel hoses.

- **Electronics:** conducting polymers in batteries, OLED displays, sensors and antistatic / EMI-shielding coatings.
- **Medicine & environment:** biodegradable PGA/PLA sutures and implants, eco-friendly compostable packaging.

QUICK RECAP

- A **polymer** is a macromolecule of many repeating **monomer** units; n = degree of polymerization.
- **Functionality** 2 → linear chain; **functionality** ≥ 3 → cross-linked network.
- **Chain-growth** (addition) is fast with no by-product; **step-growth** (condensation) is slow and releases a small molecule; **coordination** (Ziegler–Natta) gives neat stereoregular chains.
- **Thermoplastics** re-soften and remould; **thermosets** set permanently hard.
- Key plastics: **PVC**, **Teflon**, **Bakelite**, **Nylon-6,6**, **carbon fibres**; elastomers: **Buna-S** & **Buna-N**.
- **Conducting polymers** need a conjugated backbone + doping; **biodegradable polymers** (PGA, PLA) break down into CO_2, H_2O and biomass.

EXAM-ORIENTED QUESTIONS

Q1. Define polymer, monomer and degree of polymerization. *[3 marks]*

Ans. A polymer is a macromolecule made of many repeating units joined by covalent bonds. A monomer is the small molecule that repeats. Degree of polymerization (n) is the number of repeat units in a chain.

Q2. What is functionality? How does it decide the structure of a polymer? [3 marks]

Ans. Functionality is the number of reactive sites on a monomer. Functionality 2 gives a linear chain; functionality 3 or more gives a branched or cross-linked network.

Q3. Distinguish between addition and condensation polymerization. [5 marks]

Ans. Addition (chain-growth) uses C=C monomers, is fast and gives no by-product (e.g. polythene). Condensation (step-growth) uses bi-functional monomers, is slow and releases a small molecule like H_2O (e.g. Nylon-6,6).

Q4. Explain the free-radical mechanism of chain-growth polymerization. [5 marks]

Ans. Initiation: an initiator (benzoyl peroxide) forms free radicals that attack a monomer. Propagation: the radical adds monomer after monomer, growing the chain fast. Termination: two chains combine or disproportionate, stopping growth.

Q5. What is coordination polymerization? Name the catalyst and give one advantage. [3 marks]

Ans. It is polymerization using a Ziegler–Natta catalyst ($TiCl_4 + Al(C_2H_5)_3$). Advantage: it gives stereoregular (isotactic), linear, high-density polymers with controlled structure, at low pressure.

Q6. Differentiate thermoplastics and thermosetting plastics. [3 marks]

Ans. Thermoplastics have linear chains, soften on heating and can be remoulded (PVC, nylon). Thermosets have a 3-D cross-linked network, set permanently hard and cannot be remoulded (Bakelite, epoxy).

Q7. Give the monomer and two uses each of PVC and Teflon. [3 marks]

Ans. PVC: monomer vinyl chloride ($CH_2=CHCl$); uses — pipes, cable insulation. Teflon (PTFE): monomer tetrafluoroethylene ($CF_2=CF_2$); uses — non-stick cookware, gaskets/seals.

Q8. How is Nylon-6,6 made and why is it called a polyamide? [3 marks]

Ans. By condensation of adipic acid (6 C) and hexamethylenediamine (6 C). The units are joined by amide (–CO–NH–) linkages, so it is a polyamide.

Q9. What are elastomers? Compare Buna-S and Buna-N. [3 marks]

Ans. Elastomers are rubber-like polymers that stretch and recover. Buna-S (butadiene + styrene) is tough and used in tyres. Buna-N (butadiene + acrylonitrile) is oil- and heat-resistant and used in fuel hoses and oil seals.

Q10. How do conducting polymers conduct electricity? [5 marks]

Ans. They have a conjugated backbone (alternating double bonds) giving delocalized π electrons. Doping (e.g. I_2 or Na) adds or removes electrons, creating mobile charge carriers (polarons) that drift along the chain → current. Example: polyacetylene.

Q11. What are biodegradable polymers? Give two examples with uses. [3 marks]

Ans. Polymers broken down by micro-organisms into CO_2, H_2O and biomass. PGA — dissolvable surgical sutures; PLA — packaging, cutlery and 3-D printing filament.